



Solving complex linear equations involving brackets

Task A: Solving equations with multiple sets of brackets

1) Fill in the blanks to complete these two equations.

a) $2(5x + 1) + 5(x - 7) = 27$

$$10x + 2 + 5x - 35 = 27$$

$$15x - 33 = 27$$

$$15x = 60$$

$$x = 4$$

b) $4(2x + 6) + 7(4 - x) = 2(2x - 7)$

$$8x + 24 + 28 - 7x = 4x - 14$$

$$x + 52 = 4x - 14$$

$$52 = 3x - 14$$

$$66 = 3x$$

$$x = 22$$

2) Each solution contains an error. Identify it and write a sentence explaining why it is wrong.

a) $2(7x + 4) + 3x = 88$

$$14x + 8 + 6x = 88$$

$$20x + 8 = 88$$

$$20x = 80$$

$$x = 4$$

You may have written:

The terms $7x$ and 4 are inside the bracket so are correctly

multiplied by 2 but, the term $3x$ sits outside the bracket on its

own so should not have been multiplied.

b) $3(2y + 7) - 5(y + 1) = 7(3y - 2)$

$$6y + 21 - 5y + 5 = 21y - 14$$

$$y + 26 = 21y - 14$$

$$26 = 20y - 14$$

$$40 = 20y$$

$$y = 2$$

The second bracket was incorrectly multiplied out.

It should multiply out as $-5(y + 1) = -5y - 5$

The term should be -5 , not $+5$

3) Solve:

a) $2(7x - 3) + 4(x + 9) = 12$

$$14x - 6 + 4x + 36 = 12$$

$$18x + 30 = 12$$

$$18x = -18$$

$$x = -1$$

b) $3(2x + 6) + 5(x + 1) = 9(x + 3)$

$$6x + 18 + 5x + 5 = 9x + 27$$

$$11x + 23 = 9x + 27$$

$$11x = 9x + 4$$

$$2x = 4$$

$$x = 2$$

c) $4(2a + 7) = 12a - 6(a + 8)$

$$8a + 28 = 12a - 6a - 48$$

$$8a + 28 = 6a - 48$$

$$2a + 28 = -48$$

$$2a = -76$$

$$a = -38$$

d) $7(3x - 4) + 8(2 - 5x) = 9(x + 6)$

$$21x - 28 + 16 - 40x = 9x + 54$$

$$-19x - 12 = 9x + 54$$

$$-12 = 28x + 54$$

$$-66 = 28x$$

$$x = -\frac{66}{28} = -\frac{33}{14}$$



Task B: Simplifying before solving

1) Solve these equations twice. Firstly by expanding first, secondly by simplifying before expanding.

Write a sentence to compare the efficiency.

$$\begin{aligned}\text{a) } 9(x+4) + 2(x+4) &= 99 \\ 9x + 36 + 2x + 8 &= 99 \\ 11x + 44 &= 99 \\ 11x &= 55 \\ x &= 5\end{aligned}$$

$$\begin{aligned}9(x+4) + 2(x+4) &= 99 \\ 11(x+4) &= 99 \\ x+4 &= 9 \\ x &= 5\end{aligned}$$

You may have written:

Simplifying identical expressions was more efficient, with simpler steps.

$$\begin{aligned}\text{b) } 7 + 5(x-3) &= 8(x-3) - 2(x-3) \\ 7 + 5x - 15 &= 8x - 24 - 2x + 6 \\ -8 + 5x &= 6x - 18 \\ 10 + 5x &= 6x \\ x &= 10\end{aligned}$$

$$\begin{aligned}7 + 5(x-3) &= 8(x-3) - 2(x-3) \\ 7 + 5(x-3) &= 6(x-3) \\ 7 &= x-3 \\ x &= 10\end{aligned}$$

You may have written:

Simplifying identical expressions was again more efficient, and also avoided a lot of manipulating negative terms.

2) Solve:

$$\begin{aligned}\text{a) } 4x + 24 + 9(x+6) &= 24 + 11(x+6) \\ 4(x+6) + 9(x+6) &= 24 + 11(x+6) \\ 13(x+6) &= 24 + 11(x+6) \\ 2(x+6) &= 24 \\ x+6 &= 12 \\ x &= 6\end{aligned}$$

$$\begin{aligned}\text{b) } 8(y+5) &= 13 - \frac{y+5}{2} \\ 16(y+5) &= 26 - (y+5) \\ 17(y+5) &= 26 \\ 17y + 85 &= 26 \\ 17y &= -59 \\ y &= -\frac{59}{17}\end{aligned}$$

